Agentic AI System Design Roadmap (10+ LPA Engineer Track)

This roadmap is designed for transitioning from MERN Stack + Backend development into modern Agentic AI Engineering. The goal is to become capable of building scalable, production-grade multi-agent AI systems with RAG, memory, observability, deployment, and security.

Phase 1 — Backend System Design Fundamentals

What To Learn:

- Client-Server Architecture
- HTTP Lifecycle
- REST API Design
- JWT Authentication
- Request/Response Flow
- API Versioning

Where To Learn:

- YouTube: Hussein Nasser
- YouTube: CodeKarle
- Docs: FastAPI Documentation
- Docs: MDN HTTP Guide

Phase 2 — Database System Design

What To Learn:

- PostgreSQL Relationships
- Indexing
- Normalization
- Redis Caching
- Connection Pooling

- Database Scaling Basics

Where To Learn:

- YouTube: ByteByteGo
- YouTube: Gaurav Sen
- Docs: PostgreSQL Documentation
- Docs: Redis Documentation

Phase 3 — Scalable Backend Architecture

What To Learn:

- Stateless Services
- Horizontal Scaling
- Load Balancing
- Background Jobs
- Task Queues
- Celery + Redis

Where To Learn:

- YouTube: ByteByteGo
- Docs: Celery Documentation
- Docs: Redis Queue

Phase 4 — RAG System Design

What To Learn:

- Embeddings
- Chunking Strategies
- Vector Databases
- Semantic Search
- Hybrid Search
- Reranking

- Context Compression

Where To Learn:

- Docs: LlamaIndex
- Docs: LangChain
- YouTube: AI Jason
- YouTube: Cole Medin

Phase 5 — Agentic AI System Design

What To Learn:

- LangGraph
- Agent Orchestration
- Supervisor Pattern
- Task Delegation
- State Management
- Tool Calling
- Memory Systems

Where To Learn:

- Docs: LangGraph
- Docs: CrewAI
- Docs: AutoGen
- YouTube: LangChain

Phase 6 — Multi-Agent Architecture

What To Learn:

- Inter-Agent Communication
- Shared Memory
- Workflow Automation
- Agent Scheduling

- Failure Recovery
- Retry Mechanisms

Where To Learn:

- Docs: CrewAI
- Docs: AutoGen
- Research: Multi-Agent Papers

Phase 7 — Streaming & Real-Time Systems

What To Learn:

- WebSockets
- SSE Streaming
- Token Streaming
- Real-Time UI Updates

Where To Learn:

- Docs: FastAPI WebSockets
- YouTube: WebSocket Tutorials

Phase 8 — Observability & Monitoring

What To Learn:

- Tracing
- Structured Logging
- Latency Tracking
- Token Usage Monitoring
- OpenTelemetry
- LangSmith
- Arize Phoenix

Where To Learn:

- Docs: LangSmith

- Docs: OpenTelemetry
- Docs: Phoenix

Phase 9 — AI Security

What To Learn:

- Prompt Injection
- Jailbreak Attacks
- OWASP LLM Top 10
- Guardrails
- Rate Limiting
- Tool Sandboxing

Where To Learn:

- OWASP LLM Top 10
- Docs: Guardrails AI
- YouTube: AI Security Talks

Phase 10 — Deployment & DevOps

What To Learn:

- Docker
- Docker Compose
- CI/CD
- GitHub Actions
- Cloud Deployment
- AWS Basics

Where To Learn:

- Docs: Docker
- Docs: GitHub Actions
- YouTube: Nana DevOps

Phase 11 — Cost Optimization

What To Learn:

- LiteLLM
- Model Routing
- Caching
- Token Tracking
- Cost Dashboards
- Fallback Models

Where To Learn:

- Docs: LiteLLM
- OpenAI Pricing Docs

Phase 12 — MCP & AI Protocols

What To Learn:

- MCP Architecture
- MCP Servers
- Tool Exposure
- A2A Basics

Where To Learn:

- Anthropic MCP Docs
- YouTube: MCP Tutorials

Final Goal

Final Capstone Project Recommendation:
Build a Production-Grade AI Interview & Career Platform with: Multi-Agent Workflows RAG Pipelines Memory Systems Streaming Responses Observability Security Hardening Cost Tracking MCP Integration Docker Deployment

Career Goal:

By the end of this roadmap, you should be able to confidently design and deploy scalable, production-grade Agentic AI systems and target AI Engineer / Backend AI roles in the 10–20 LPA range.